

Dynamics and Processes on Complex Networks (DPCN)

Assignment 1: Opinion Network Formation & Structural Analysis

Quantitative Modeling of Ideological Alignment, Thematic Communities, and Percolation Transitions

Team Name: Lost in the Network

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1 Team Information

- **Team Name:** Lost in the Network
- **Group Members and Roll Numbers:**
 1. **Ampilli Sai Venkata Naga Bhargav** — Roll Number: 2023112014
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- **Course:** Dynamics and Processes on Complex Networks (DPCN)

2 GitHub Link to Code Repository

The complete source code, preprocessing modules, network construction algorithms, high-resolution figures, and reproducible execution pipelines are publicly hosted at:

https://github.com/saiXbhargav/DPCN_assignment1

The repository is organized modularly: pipeline execution is automated by `run_pipeline.py`, core algorithms reside in `src/`, raw survey responses are provided in `Survey_Results_UC.csv`, and all verified outputs and publication figures are generated into `output/`.

3 Dataset Documentation

3.1 Survey Design and Domain Modularization

The empirical dataset (`Survey_Results_UC.csv`) comprises responses collected from an undergraduate student cohort across **60 standardized opinion statements**. The questionnaire is structured into four balanced thematic modules of 15 questions each:

1. **Technology (T01–T15):** Societal impacts of artificial intelligence, generative AI adoption in higher education, algorithmic transparency, automated decision-making ethics, autonomous mobility, digital surveillance, and healthcare diagnostic automation.
2. **Education (E01–E15):** Project-based learning paradigms, traditional written examinations, compulsory classroom attendance policies, online instructional delivery, continuous evaluation mechanisms,

multidisciplinary curricula, and undergraduate research mandates.

3. **Society & Ethics (S01–S15):** Social media disinformation dynamics, mandatory university community service, digital privacy rights, data sovereignty, algorithmic accountability in public governance, free expression boundaries, and inclusive academic discourse.
4. **Environment (V01–V15):** Climate urgency prioritization, renewable energy transition investments, institutional waste segregation, circular economy practices, corporate environmental liability, biodiversity preservation, and intergenerational ecological ethics.

3.2 Data Hygiene and Quality Control

The raw survey file contains 96 rows and 61 columns, with the initial column serving as a unique respondent identifier (`id. Response ID`) spanning non-consecutive indices from 1 to 125. A programmatic audit confirmed zero duplicate identifiers and zero redundant rows.

To guarantee mathematical consistency across vector representations, a two-stage quality-control protocol was implemented:

- **Vacant Row Elimination:** Exactly 5 rows were completely blank across all 60 survey statements (IDs 44, 60, 68, 73, 78). These empty records were purged.
- **Incomplete Response Filtering:** Exactly 4 respondents (IDs 30, 39, 77, 87) completed only the initial 15 Technology questions and abandoned all remaining 45 statements. Because pairwise vector correlations require sufficient overlapping dimensionality to prevent variance distortion, respondents completing fewer than 30 statements were removed.
- **Retained Cohort ($N = 87$):** Exactly $96 - 5 - 4 = 87$ active respondents were retained in the clean cohort (85 respondents completed all 60 items, 1 completed 50 items, and 1 completed 45 items).

3.3 Likert Scale Encoding and Imputation

Qualitative 5-point Likert survey responses were mapped onto a bipolar zero-centered interval scale

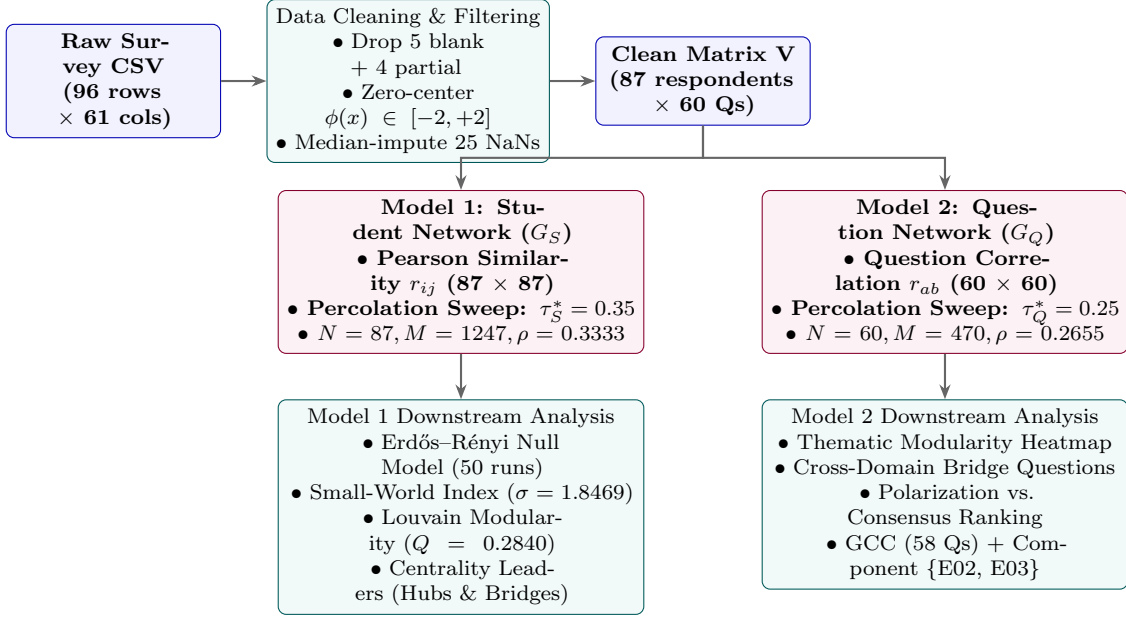


Figure 1: End-to-end analytical execution pipeline illustrating survey data ingestion, quality filtering, zero-centered Likert encoding, dual-network construction, percolation sweeps, null-model benchmarking, and downstream topological analysis.

$\phi(x) \in \{-2, -1, 0, 1, 2\}$:

$$\phi(x) = \begin{cases} +2 & \text{Strongly Agree} \\ +1 & \text{Agree} \\ 0 & \text{Neutral / No Comments} \\ -1 & \text{Disagree} \\ -2 & \text{Strongly Disagree} \end{cases} \quad (1)$$

Handling Missing Values vs. Neutral Entries: Across the active $87 \times 60 = 5,220$ response matrix, exactly **37 cells** contained explicit “No Comments” entries, which were encoded directly as 0 (neutral). Exactly **25 cells** (0.479% of active data) were blank (NaN). These 25 missing entries were imputed using the column-wise question median score (`df.numeric.fillna(medians)`), preserving central tendency metrics without artificially inflating sample variance.

Scale Invariance of Pearson Correlation: Pearson correlation r_{ij} is mathematically invariant under positive linear affine transformations ($v \mapsto a \cdot v + b, a > 0$). Centering Likert scores from $\{1, \dots, 5\}$ to $\{-2, \dots, +2\}$ does not alter Pearson correlation values because individual means $\bar{v}_i$ are subtracted during standard covariance normalization. Zero-centering provides an intuitive symmetric baseline for evaluating domain consensus and profiling community personas.

4 Pipeline Followed: Network Formulations and Modeling

4.1 Execution Workflow

The analysis constructs and evaluates two complementary network architectures driven by `run_pipeline.py`,

as illustrated in Figure 1:

1. **Model 1: Student Opinion Network (G_S):** $N = 87$ active student nodes; edges denote pairwise profile alignment across all 60 questions.
2. **Model 2: Question Concept Network (G_Q):** $N_Q = 60$ question nodes; edges denote cross-cohort co-variation across all 87 respondents.

4.2 Mathematical Network Formulations

4.2.1 Model 1: Student Opinion Network (G_S)

Let $V_S = \{s_1, \dots, s_N\}$ denote the $N = 87$ students, where each student i is characterized by an opinion vector $\mathbf{v}_i \in \mathbb{R}^K$ ($K = 60$). Pairwise alignment is quantified via the sample **Pearson Correlation Coefficient**:

$$r_{ij} = \frac{\sum_{k=1}^K (v_{ik} - \bar{v}_i)(v_{jk} - \bar{v}_j)}{\sqrt{\sum_{k=1}^K (v_{ik} - \bar{v}_i)^2} \sqrt{\sum_{k=1}^K (v_{jk} - \bar{v}_j)^2}} \in [-1, 1] \quad (2)$$

where $\bar{v}_i = \frac{1}{K} \sum_{k=1}^K v_{ik}$. Applying cutoff τ_S^* , binary adjacency matrix A and weighted matrix W are defined as:

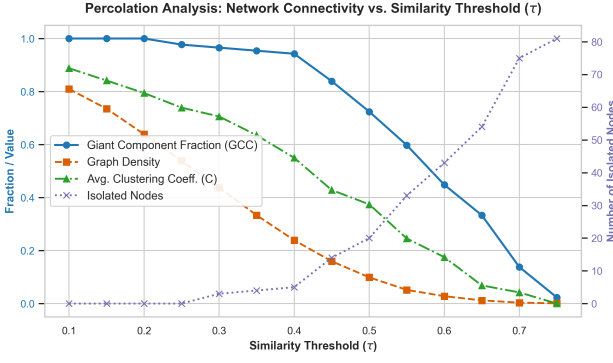
$$A_{ij} = \begin{cases} 1 & \text{if } r_{ij} \geq \tau_S^* \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

$$W_{ij} = \begin{cases} r_{ij} & \text{if } r_{ij} \geq \tau_S^* \text{ and } i \neq j \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

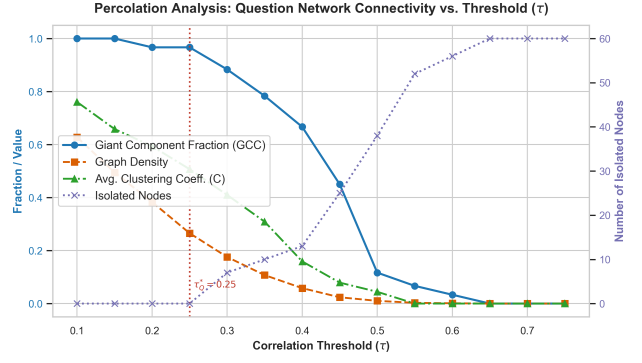
Self-loops are excluded ($A_{ii} = 0$). Negative correlations and sub-threshold positive alignments are mapped to non-edges.

4.2.2 Model 2: Question Concept Network (G_Q)

Let $V_Q = \{q_1, \dots, q_K\}$ denote the 60 questions, represented by column vectors $\mathbf{q}_a \in \mathbb{R}^N$. Pairwise correla-



(a) Model 1: Student Network percolation sweep.



(b) Model 2: Question Network percolation sweep.

Figure 2: Percolation threshold parameter sweeps across $\tau \in [0.10, 0.75]$. Primary axes display GCC fraction, graph density, and average clustering coefficient; secondary axes display isolated node count. Selected cutoffs $\tau_S^* = 0.35$ and $\tau_Q^* = 0.25$ maximize network sparsity while preserving $\geq 95\%$ giant-component connectivity.

tion is $r_{q_a, q_b} = \text{corr}(\mathbf{q}_a, \mathbf{q}_b)$. Edges exist if $r_{q_a, q_b} \geq \tau_Q^*$ ($a \neq b$), weighted by $W_{ab} = r_{q_a, q_b}$.

4.3 Percolation Threshold Analysis

We performed systematic percolation sweeps across $\tau \in [0.10, 0.75]$ with step size $\Delta\tau = 0.05$, tracking edges (M), density (ρ), Giant Connected Component fraction ($|V_{\text{GCC}}|/N$), isolates, and clustering (C).

Selection Rule: Select the largest threshold τ^* that preserves $\geq 95\%$ of nodes in the Giant Connected Component.

- **Student Network (G_S):** As shown in Figure 2a, GCC coverage remains robust at 95.40% (83/87 nodes, 4 isolates) at $\tau = 0.35$, while density drops to $\rho = 0.3333$ ($M = 1,247$, filtering 58.8% of edges from $\tau = 0.10$). Above $\tau = 0.40$, fragmentation accelerates (GCC drops to 83.91% at $\tau = 0.45$; isolates jump to 14). Thus, $\tau_S^* = 0.35$ is selected.
- **Question Network (G_Q):** As shown in Figure 2b, GCC coverage holds at 96.67% (58/60 nodes) through $\tau = 0.25$ before falling to 88.33% at $\tau = 0.30$ and 78.33% at $\tau = 0.35$. Thus, $\tau_Q^* = 0.25$ ($M = 470, \rho = 0.2655$) is the percolation-justified threshold.

4.4 Topological Metrics and Null Model Benchmarking

We compute standard network metrics:

- **Degree Centrality (C_D):** $C_D(i) = \frac{k_i}{N-1} = \frac{1}{N-1} \sum_{j \neq i} A_{ij}$.
- **Betweenness Centrality (C_B):** $C_B(i) = \frac{\sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}}{\sigma_{st}}$.
- **Closeness Centrality (C_C):** $C_C(i) = \frac{N-1}{\sum_{j \neq i} d(i, j)}$.
- **Local Clustering (C_i):** $C_i = \frac{2e_i}{k_i(k_i-1)} = \frac{\sum_{j, k} A_{ij} A_{jk} A_{ki}}{k_i(k_i-1)}$.
- **Erdős–Rényi $G(N, p)$ Baseline:** Benchmark G_S against an ensemble of 50 simulated random graphs ($N = 87, p = 0.3333$).
- **Small-World Index (σ):** Following Watts-Strogatz

and Humphries-Gurney:

$$\gamma = \frac{C_{\text{empirical}}}{C_{\text{ER}}^{\text{simulated}}}, \quad \lambda = \frac{L_{\text{empirical}}}{L_{\text{ER}}^{\text{simulated}}}, \quad \sigma = \frac{\gamma}{\lambda} \quad (5)$$

A network exhibits small-world organization when $\gamma > 1$ and $\lambda \approx 1$, yielding $\sigma > 1$.

5 Analysis and Visualizations

5.1 Global Properties and Random Benchmarking

Table 1 compares G_S against the 50-run Erdős–Rényi ensemble.

Key measured findings include:

- **Clustering Elevation ($\gamma = 1.9015$):** The empirical clustering ($C = 0.6350$) is nearly double the random null model ($C_{\text{ER}} = 0.3340 \pm 0.0077, z \approx 39.1$).
- **Geodesic Path Parity ($\lambda = 1.0295$):** Empirical GCC path length ($L = 1.7161$) matches the random baseline ($L_{\text{ER}} = 1.6669 \pm 0.0074$) within 3.0%.
- **Small-World Index ($\sigma = 1.8469$):** With $\gamma = 1.9015 > 1$ and $\lambda = 1.0295 \approx 1$, $\sigma = 1.8469$ confirms small-world properties relative to ER random graphs.
- **Degree Assortativity ($r = -0.0199$):** Near-zero correlation indicates that consensus hubs connect broadly across high- and low-degree peers rather than forming an isolated core.

5.2 Degree Distribution vs. Poisson Null Model

Figure 3 contrasts empirical degrees against the theoretical Poisson fit ($\lambda = 28.7$).

The empirical distribution displays high variance (≈ 208.4) compared to the Poisson baseline (variance = $\lambda = 28.7$). The right-skewed tail reflects consensus hubs connected to $> 65\%$ of peers, alongside peripheral isolates.

Table 1: Global topological properties of the empirical Student Opinion Network (G_S) at percolation threshold $\tau_S^* = 0.35$, benchmarked against an ensemble of 50 Erdős–Rényi $G(N, p)$ random graphs and theoretical expectations.

Metric	Empirical Network (G_S)	ER Baseline (Simulated, $N = 50$)	Theoretical ER	Ratio / Significance
Number of Nodes (N)	87	87	87	Exact match
Number of Edges (M)	1,247	$1,247 \pm 28.5$	—	Exact match
Average Degree ($\langle k \rangle$)	28.667	28.667 ± 0.655	$p(N-1) = 28.667$	Exact match
Graph Density (ρ)	0.3333	0.3333 ± 0.0076	$p = 0.3333$	Exact match
Clustering Coefficient (C)	0.6350	0.3340 $\pm$ 0.0077	$p = 0.3333$	$\gamma = C/C_{ER} = 1.9015$ ($z \approx 39.1$)
Network Transitivity	0.6500	0.3339 ± 0.0075	$p = 0.3333$	$1.9467 \times$ ER baseline
GCC Node Count	83 (95.40%)	87.0 ± 0.0 (100%)	—	4 peripheral isolates
Connected Components	5	1.0 ± 0.0	—	1 GCC + 4 singletons
Avg. Path Length GCC (L)	1.7161	1.6669 $\pm$ 0.0074	$\frac{\ln N}{\ln \langle k \rangle} = 1.3308$	$\lambda = L/L_{ER} = 1.0295$
Graph Diameter (GCC)	4	2.0 ± 0.0	—	Worst-case geodesic
Degree Assortativity (r)	-0.0199	0.0012 ± 0.0241	0.0	Weakly disassortative
Louvain Modularity (Q)	0.2840	—	—	4 primary communities
Small-World Index (σ)	1.8469	1.0000	1.0000	$\sigma = \gamma/\lambda > 1$

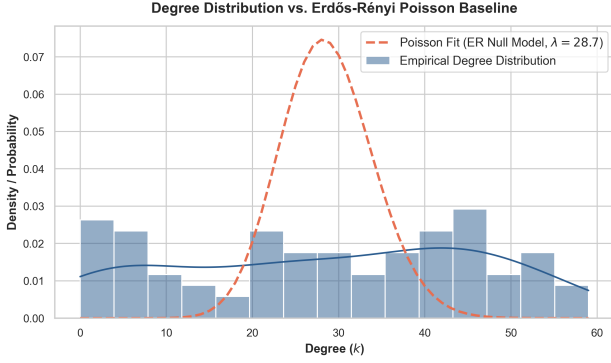


Figure 3: Degree distribution of G_S vs. theoretical Poisson fit ($\lambda = 28.7$) of the Erdős–Rényi baseline, showing high variance and positive skew.

5.3 Consensus Hubs vs. Structural Bridges

Table 2 lists the top ranking respondents by Degree (C_D) and Betweenness (C_B).

- **Consensus Hubs:** Student 114 achieves the highest degree ($k = 59, C_D = 0.6860$, Closeness 0.7446), followed by Student 55 ($k = 58$) and Student 61 ($k = 56$).
- **Topological Bridges:** Student 90 exhibits the highest betweenness ($C_B = 0.0346, k = 52$), followed by Student 19 ($C_B = 0.0307, k = 41$) and Student 53 ($C_B = 0.0290, k = 48$). Student 114 ranks fourth in betweenness (0.0288) while leading in degree.

5.4 Network Topology and Correlation Structure

Figure 4a illustrates the force-directed layout of G_S , colored by Louvain community with node sizes scaled by betweenness. The four dominant communities form cohesive clusters surrounding central bridge nodes.

Figure 4b displays the 60×60 question correlation matrix. Strong positive correlation blocks appear in **Environment** (V) and **Society** (S), while **Education** (E) exhibits lower internal correlation, reflecting heterogeneous sub-topics.

Table 2: Top ranking respondents in the Student Opinion Network (G_S) classified by Degree Centrality (consensus hubs) and Betweenness Centrality (topological bridges).

ID	Degree	C_D	Strength	C_B	C_C	Comm.
Top 5 Consensus Hubs (Degree Centrality)						
114	59	0.6860	30.327	0.0288	0.7446	Comm 3
55	58	0.6744	28.125	0.0192	0.7376	Comm 1
61	56	0.6512	28.831	0.0164	0.7173	Comm 4
49	55	0.6395	28.448	0.0166	0.7173	Comm 4
70	55	0.6395	26.564	0.0255	0.7108	Comm 1
Top 5 Topological Bridges (Betweenness Centrality)						
90	52	0.6047	26.466	0.0346	0.6919	Comm 2
19	41	0.4767	18.260	0.0307	0.6305	Comm 1
53	48	0.5581	22.841	0.0290	0.6626	Comm 2
114	59	0.6860	30.327	0.0288	0.7446	Comm 3
41	35	0.4070	16.148	0.0264	0.5923	Comm 2

6 Results and Discussion

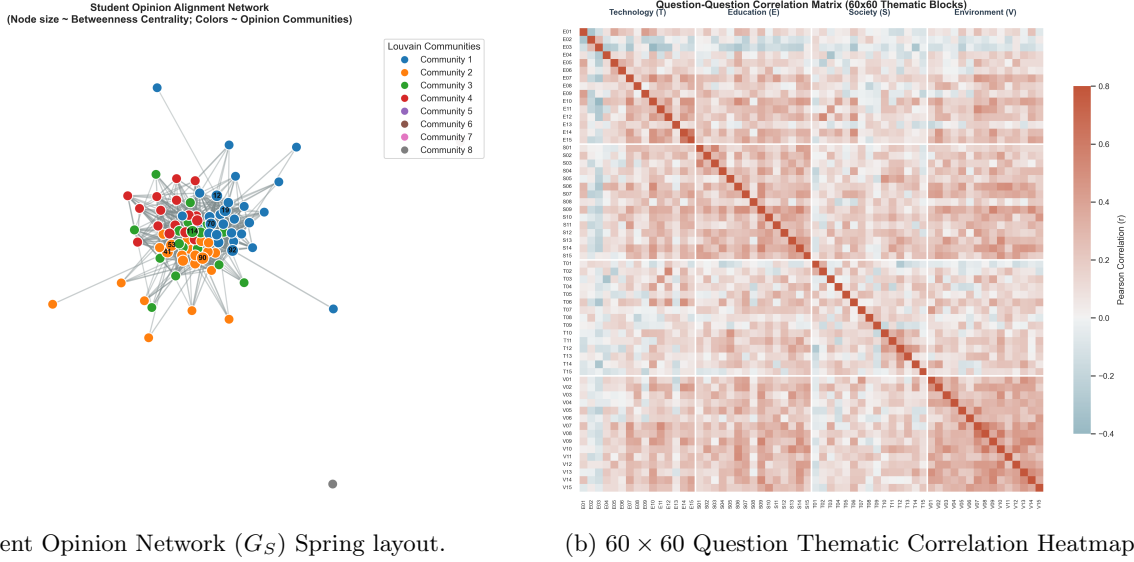
6.1 Community Structure and Thematic Personas

Applying weighted Louvain modularity optimization on G_S (seed=42) yields modularity $Q = 0.2840$, partitioning the cohort into **four primary opinion communities** (95.40% of students) plus **four singleton isolates** (Table 3 and Figure 5a).

To establish a comprehensive understanding of opinion organization, we conduct a multi-scale analysis combining **group-level community opinion profiles** (Figure 5a) with **item-level question response dispersion** (Figure 5b). While community profiling identifies how structurally partitioned sub-cohorts systematically diverge in their domain-level philosophies (e.g., Techno-Optimists vs. Eco-Centric Reformers), question-wise variance isolates the specific pedagogical and socio-technical policy statements that trigger polarization versus universal consensus across the student body.

Ideological Personas:

1. **Community 1: Progressive Institutionalists** ($N = 28, 32.2\%$): Balanced stance (+1.120 overall), supporting environmental mandates (+1.369) and structured educational reform (+1.029).
2. **Community 2: Civic Champions** ($N =$



(a) Student Opinion Network (G_S) Spring layout.

(b) 60×60 Question Thematic Correlation Heatmap.

Figure 4: Visual topology: (a) Student Opinion Network ($N = 87, \tau_S^* = 0.35$) with nodes colored by Louvain community and sized by betweenness centrality; (b) Symmetric 60×60 correlation matrix partitioned into 15-question thematic blocks (T, E, S, V).

Table 3: Louvain community structure ($Q = 0.2840$) and thematic mean response profiles across the four survey domains (Scale: -2.0 to $+2.0$).

Community ID	Size (N)	Share (%)	Tech (T)	Edu (E)	Society (S)	Env (V)	Overall Stance	Ideological Persona Characterization
1	28	32.2%	+0.874	+1.029	+1.207	+1.369	+1.120	Progressive Institutionalists: Balanced; pro-ethics/environment. Civic & Environmental Champions: Highest overall agreement. Techno-Optimists: Strong AI endorsement; critical of edu mandates. Eco-Centric Reformers: Highest environmental priority.
2	22	25.3%	+1.036	+1.033	+1.473	+1.412	+1.239	
3	17	19.5%	+1.173	+0.773	+1.192	+1.455	+1.148	
4	16	18.4%	+0.996	+0.787	+1.350	+1.508	+1.160	
5 (ID 40)	1	1.1%	+1.133	+1.133	+1.667	+1.733	+1.417	Singleton Isolate: Uniformly high positive responses.
6 (ID 119)	1	1.1%	+0.800	0.000	+0.067	+0.200	+0.267	Singleton Isolate: Neutral on education and society.
7 (ID 120)	1	1.1%	-0.400	-0.133	0.000	-0.267	-0.200	Singleton Isolate: Cohort dissenter (negative overall stance).
8 (ID 125)	1	1.1%	+1.200	+0.333	+0.667	+0.533	+0.683	Singleton Isolate: Moderate idiosyncratic profile.

22, 25.3%): Highest overall agreement (+1.239), with strong conviction in ethics (+1.473) and sustainability (+1.412).

- Community 3: Techno-Optimists** ($N = 17, 19.5\%$): Strongest AI endorsement (+1.173), but critical of educational mandates (+0.773).
- Community 4: Eco-Centric Reformers** ($N = 16, 18.4\%$): Highest environmental conviction (+1.508), prioritizing sustainability over tech disruption (+0.996).
- Peripheral Isolates (Comm 5–8, 1 student each)**: Student 40 (uniformly agreeable, +1.417); Student 119 (neutral on education 0.000); Student 120 (lone dissenter, -0.200 overall); Student 125 (moderate, +0.683).

6.2 Path Length Across Canonical Topologies

Table 4 benchmarks G_S against extreme graph topologies ($N = 87$).

Table 4: Comparison of shortest path length and diameter against canonical graph topologies ($N = 87$).

Topology	Diameter	Avg. Path Length (L)	Structural Basis
Chain (Path Graph)	86	29.333	$L = (N + 1)/3$
Star Graph	2	1.977	$L = 2(N - 1)/N$
Erdős-Rényi (Sim.)	2	1.667 ± 0.007	Mean over 50 runs ($p = 0.3333$)
Empirical Network (G_S)	4	1.716	Observed GCC ($\tau_S^* = 0.35$)

Empirical path length (1.716) is over $17\times$ shorter than a linear chain (29.333) and approaches the star graph limit (1.977). Unlike a star graph with a single vulnerable hub, the network achieves reachability via a **distributed core of consensus hubs** (IDs 114, 55, 61).

The empirical diameter ($\text{diam} = 4$) is twice the random baseline ($\text{diam} = 2$), reflecting **community modularity** ($Q = 0.2840$): traversing between peripheral nodes across communities requires routing through intermediate bridge students (IDs 90, 19, 53).

6.3 Question Polarization and Consensus Analysis

While the Louvain community analysis in Section 6.1 reveals macro-level divergence across opinion personas, Figure 5b details the complementary item-level response variance (σ^2) across all 60 survey questions, sorted in ascending order of dispersion. Table 5 complements this visual distribution with exact numerical centralities and statement texts for the extreme consensus and divisive items.

As visualized in Figure 5b, response variance spans a five-fold range from $\sigma^2 = 0.254$ to $\sigma^2 = 1.365$ across the 60 items (mean variance $\mu_{\sigma^2} = 0.72$). Crucially, opinion dispersion is not uniformly distributed across the four domains, but concentrates heavily within institutional educational policies.

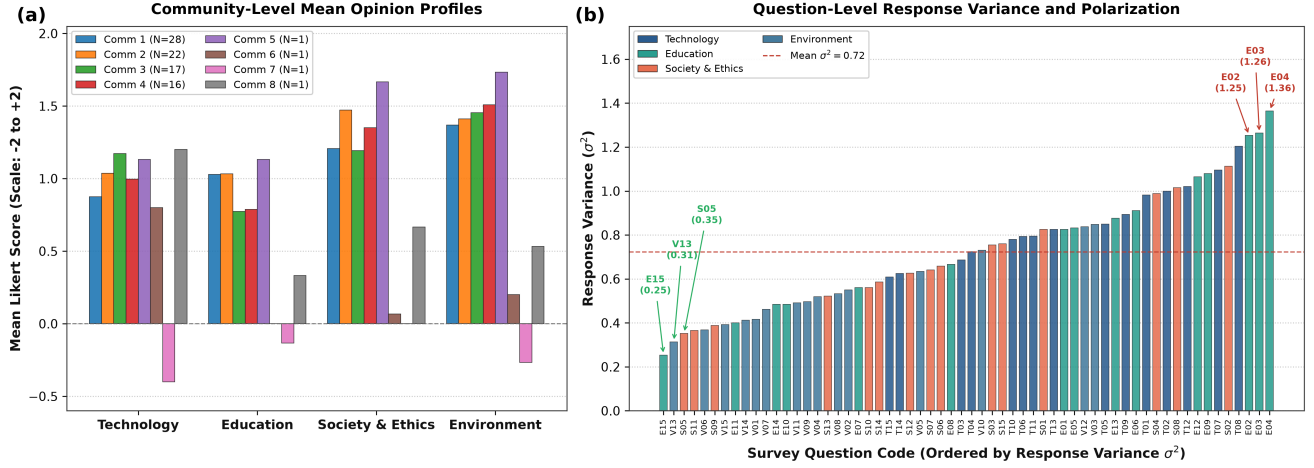


Figure 5: Community-level and question-level opinion structure. (a) Mean Likert response profiles across Technology, Education, Society & Ethics, and Environment for the Louvain communities identified in the Student Opinion Network. (b) Question-wise response variance across all 60 survey questions (ordered by variance), with higher variance indicating greater disagreement within the cohort. The most divisive questions are concentrated in the Education domain, while the lowest-variance questions indicate stronger cohort consensus.

Table 5: Polarization and consensus analysis: Top 3 most divisive questions (highest response variance σ^2) contrasted with top 3 highest consensus questions (lowest response variance and high mean) across the 60 survey items.

Rank	Code	Category	Variance (σ^2)	Mean Score	Betweenness (C_B)	Survey Statement Text
Panel A: Top Divisive / Polarizing Questions						
1	E04	Education	1.365	+0.437	0.0000	High-quality online learning can effectively complement classroom teaching.
2	E03	Education	1.264	-0.943	0.0000	Class attendance should be compulsory for all courses.
3	E02	Education	1.253	-0.161	0.0000	Traditional written examinations accurately measure a student's knowledge.
Panel B: Top Consensus / Convergent Questions						
1	E15	Education	0.254	+1.713	0.0075	Continuous learning and skill development are essential throughout one's career.
2	V13	Environment	0.314	+1.678	0.0067	Companies should be held accountable for the environmental impacts of their activities.
3	S05	Society & Ethics	0.352	+1.632	0.0019	Individuals should have greater control over how their personal data are collected and used.

The Education Policy Fault Line: The top 3 most divisive statements across the entire survey all originate in **Education**: **E04** (*online learning effectiveness*, $\sigma^2 = 1.365$), **E03** (*compulsory class attendance*, $\sigma^2 = 1.264$), and **E02** (*written examination efficacy*, $\sigma^2 = 1.253$). This item-level polarization directly underpins the community-level divergence observed in Figure 5a, where Community 3 (Techno-Optimists, Edu mean +0.773) and Community 4 (Eco-Centric Reformers, Edu mean +0.787) exhibit substantially lower education support than Communities 1 and 2 (+1.029 and +1.033).

Universal Cohort Consensus: Conversely, the consensus tail in Figure 5b demonstrates that the cohort converges with strong unanimity on foundational principles: continuous learning and skill development (**E15**, $\sigma^2 = 0.254$, Mean +1.713), corporate environmental accountability (**V13**, $\sigma^2 = 0.314$, Mean +1.678), and personal data sovereignty (**S05**, $\sigma^2 = 0.352$, Mean +1.632). This explains why all four Louvain communities maintain uniformly positive and elevated mean scores in Environment (+1.369 to +1.508) and Society (+1.192

to +1.473): ideological fragmentation is confined to pedagogical methods rather than environmental or ethical mandates.

6.4 Question Concept Network and Thematic Bridges

Figure 6 visualizes G_Q at $\tau_Q^* = 0.25$. The network forms a 58-node Giant Component alongside a disconnected 2-node component $\{\mathbf{E02}, \mathbf{E03}\}$, which correlate together ($r \geq 0.25$) but detach from the rest of the concept graph.

Top cross-thematic bridge questions by betweenness centrality:

1. **S07** ($C_B = 0.0563$, **Deg 29**): *Equal opportunities regardless of background.*
2. **E10** ($C_B = 0.0504$, **Deg 31**): *Innovation and problem-solving over rote learning.*
3. **V11** ($C_B = 0.0460$, **Deg 22**): *Technology essential for environmental challenges.*
4. **S15** ($C_B = 0.0450$, **Deg 27**): *Public trust for responsible emerging technologies.*

These findings demonstrate that cross-domain connectivity is anchored by **equity** (**S07**), **pedagogical reform** (**E10**), and the **technology-environment**

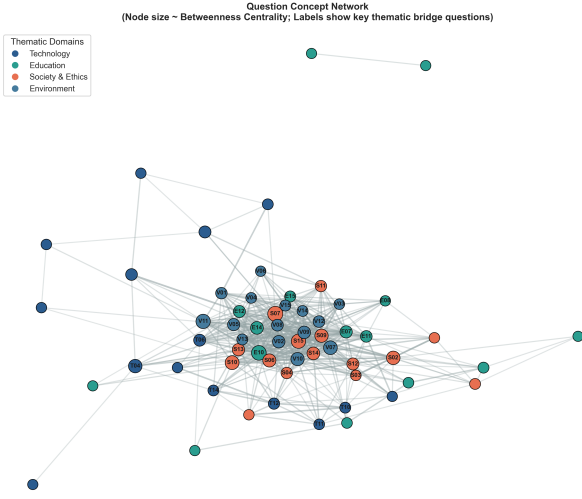


Figure 6: Question Concept Network (G_Q at $\tau_Q^* = 0.25$, $N = 60$, $M = 470$). Node colors denote domains; node sizes scale with Betweenness Centrality.

nexus (V11).

6.5 Methodological Limitations

1. **Static Response Topology:** Edges denote statistical alignment in survey response space rather than direct interpersonal communication, friendship, or social influence. While correlation networks accurately map ideological cohesion, they do not capture dynamic opinion propagation over time.
2. **Degree-Unmatched Null Model:** The Erdős–Rényi random baseline $G(N, p)$ assumes independent edge probabilities and a Poisson degree distribution, which does not preserve the empirical degree sequence. A degree-preserving configuration model or Chung–Lu model would provide an even more stringent benchmark for small-world validation.
3. **Interval Assumption on Ordinal Data:** Pearson correlation assumes equal-interval distances between discrete Likert response levels. Although zero-centered scaling preserves symmetry, future studies could explore polychoric correlation or distance metric learning tailored to ordinal scales.
4. **Sample Specificity and Generalizability:** The empirical dataset reflects a specific cohort of undergraduate engineering students ($N = 87$). The observed high consensus on sustainability and data privacy may reflect domain-specific training rather than broader societal distributions.

7 Individual Contribution

All team members contributed equitably (33.3% each) to the conceptualization, mathematical formulation, code implementation, figure generation, and report writing for this assignment. The individual task allocation and verified contribution breakdown are structured below:

Ampilli Sai Venkata Naga Bhargav

(2023112014): Erdős–Rényi null model simulation ensemble ($N = 50$), Small-World Index

computation, zero-centered Likert scale mathematical encoding, Model 1 percolation threshold sweep implementation, and Louvain community detection algorithm formulation.

Chaitanya Thogata (2023113018): Model 1 network construction, Pearson similarity formulation, node centrality computations (Degree, Betweenness, Closeness), and consensus hub vs. topological bridge comparative analysis.

Pasumarthy Deepak (2023101109): missing-value imputation strategy (median), Survey data ingestion, Model 2 (Question Concept Network and percolation sweep), generation of publication figures, and comprehensive report compilation.

8 Conclusion

In this study, we transformed high-dimensional survey responses into dual complex networks to analyze opinion organization across Technology, Education, Society, and Environment. Through quality filtering ($N = 87$) and percolation thresholding ($\tau_S^* = 0.35$, $\tau_Q^* = 0.25$), we established that the Student Opinion Network exhibits structural properties of **small-world organization** ($\sigma = 1.8469$), combining elevated clustering ($\gamma = 1.9015$) with short geodesic paths ($\lambda = 1.0295$). Weighted Louvain community detection ($Q = 0.2840$) identified four primary ideological cohorts connected via a distributed core of consensus hubs and bridge students. Question-level analysis revealed that institutional educational policies constitute the primary axis of cohort polarization, while inter-domain conceptual connectivity is sustained by foundational principles of social equity, educational innovation, and sustainable technology.